

## Homework (Carolina Reaper)

**Q1.** Solve for  $x$ :  $2^x = 32$

- (A)  $x = 3$
- (B)  $x = 4$
- (C)  $x = 5$
- (D)  $x = 6$
- (E)  $x = 7$

**Q2.** A music producer creates a rhythm pattern with a frequency of  $3^x$  Hz. If  $x = 2$ , what is the frequency?

- (A) 9 Hz
- (B) 12 Hz
- (C) 15 Hz
- (D) 18 Hz
- (E) 20 Hz

**Q3.** An artist creates a canvas with a size of  $2^x \cdot 3^y$  square inches. If  $x = 3$  and  $y = 2$ , what is the size?

- (A) 72 sq in
- (B) 108 sq in
- (C) 144 sq in
- (D) 180 sq in
- (E) 216 sq in

**Q4.** Solve for  $y$ :  $5^y = 125$

- (A)  $y = 1$
- (B)  $y = 2$
- (C)  $y = 3$
- (D)  $y = 4$
- (E)  $y = 5$

**Q5.** A sound engineer adjusts the gain of a microphone to  $4^x$  dB. If  $x = 1.5$ , what is the gain?

- (A) 6 dB
- (B) 8 dB
- (C) 10 dB
- (D) 12 dB
- (E) 16 dB

**Q6.** Solve for  $z$ :  $2^z = 64$

- (A)  $z = 4$
- (B)  $z = 5$
- (C)  $z = 6$
- (D)  $z = 7$
- (E)  $z = 8$

**Q7.** A painter uses a color with a hue of  $3^x$  degrees. If  $x = 1$ , what is the hue?

- (A) 3 deg
- (B) 6 deg
- (C) 9 deg
- (D) 12 deg
- (E) 15 deg

**Q8.** Solve for  $w$ :  $5^w = 25$

- (A)  $w = 1$
- (B)  $w = 1.5$
- (C)  $w = 2$
- (D)  $w = 2.5$
- (E)  $w = 3$

### Open Ended Questions (FRQ)

**Q9.** A musician creates a song with a tempo of  $2^x$  beats per minute. If the tempo increases by a factor of  $3^y$ , what is the new tempo in terms of  $x$  and  $y$ ?

**Q10.** An artist designs a mosaic with a pattern of  $4^x \cdot 2^y$  tiles. If  $x = 2$  and  $y = 1$ , what is the total number of tiles used?

### Chef's Tips

- Check calculations carefully
- Consider multiple approaches to solve problems
- Use mathematical notation correctly